

```
import pandas as pd
df = pd.read_csv("messy_sales.csv")
df.shape
df.head(10)
df.dtypes
df.describe
```

#I see that it shows NaN for prices. Also the dates are shown in a different way for some o
#There is also a negative for quantity which does not make sense.

```
df.isna().sum()
```

```
fill_value = df["price"].median()
df["price"] = df["price"].fillna(fill_value)
```

```
df.isna().sum()
```

```
df.shape
```

```
print(df.duplicated().sum())
df = df.drop_duplicates()
```

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```
df.shape
```

```
df["zip"] = df["zip"].astype(str).str.zfill(5)
```

```
df["date"] = pd.to_datetime(df["date"], format="mixed")
```

```
df.shape
df.head(10)
df.dtypes
df.describe
```

```
df[df["qty"] < 0]
```

```
df.to_csv("sales_clean.csv", index=False)
```

```
## Cleaning Log
```

```
#I checked for duplicates and took those out, leaving 292 rows
```

```
#There were also 12 missing prices
```

```
#I decided to go with the median to keep the data as unbiased as I can
```

```
#The zipcodes were stored as numbers so the ones leading with 0 were dropped
```

```
#I changed them into text so the full 5 digits would show up.
```

#The dates were shown in different formats
#I standardized them with pd.to_datetime

#There were quantites with a negative number

#I dropped those because there is no way to verify **if** they were returns or positive numbers

#It made it more simple to just drop them instead of counting them as valid rows.

#The wrong choice here could mislead a business by showing wrong information, I think that
#having no information.